

Introduction

Problem Statement

Smallholder farmers frequently lack access to agronomic expertise when deciding which crop to cultivate. Selecting an unsuitable crop given local soil and climate conditions leads to reduced yields, resource waste, and economic loss. This project addresses that gap by building a machine learning model that recommends the most appropriate crop based on measurable soil and environmental parameters. The problem is formulated as a 22-class classification task: given seven numerical features (Nitrogen, Phosphorus, Potassium, soil pH, temperature, humidity, and rainfall), the model predicts the optimal crop from a set of 22 species. Such a decision-support tool has direct practical value for extension services and precision-agriculture applications.

Dataset Description

We will use the publicly available Crop Recommendation Dataset from Kaggle (<https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>). The dataset contains 2,200 samples with 100 observations per crop class, making it balanced across the 22 target categories. Each sample includes seven continuous numerical features: nitrogen (N), phosphorus (P), potassium (K), soil pH, temperature (°C), relative humidity (%), and rainfall (mm). The dataset will be downloaded from Kaggle and stored directly in the project repository.

Initial Cleaning and Feature Engineering:

1. **Missing Values Check:** An initial check confirmed the absence of any missing values across all features, eliminating the need for imputation or removal of incomplete records.
2. **Feature-Target Separation:** The dataset was separated into features (X) and the target variable (y). The label column was designated as y, while the remaining seven numerical columns formed X.

3. **Feature Scaling (StandardScaler):** Given the varied ranges and distributions of the numerical features, StandardScaler was applied to X. This preprocessing step transformed the features to have a mean of 0 and a standard deviation of 1, preventing features with larger numerical ranges from disproportionately influencing the machine learning models. The scaled features were stored in a new DataFrame, X_scaled, which is ready for model training.

Data Organization

To ensure robust model evaluation and prevent overfitting, the preprocessed data (features X_scaled and target y) was split into training and test sets. An 80/20 split was utilized, allocating 80% of the data for model training and 20% for evaluating the trained models' performance on unseen data.

Crucially, stratified splitting was used during this process. This approach ensures that the proportion of each of the 22 crop classes is maintained across both the training and test sets. Stratification is particularly important in multi-class classification tasks with balanced datasets (as observed in the EDA), as it helps to prevent any single class from being underrepresented or overrepresented in either set, thereby ensuring that the models are trained and evaluated on a representative distribution of crop types.

Methods

To develop a robust crop recommendation system, three distinct machine learning classification algorithms were used and rigorously tuned: Decision Tree, Random Forest, and K-Nearest Neighbors.

Machine Learning Models

- **Decision Tree Classifier:** This model makes predictions by learning simple decision rules inferred from the data features. It partitions the data into subsets based on feature values, forming a tree-like structure of decisions. Key hyperparameters tuned included max_depth (the maximum depth of the tree), min_samples_leaf (the minimum number of samples required to be at a leaf node), min_samples_split (the minimum number of samples required to

split an internal node), and criterion (the function to measure the quality of a split, 'gini' or 'entropy').

- **Random Forest Classifier:** An ensemble learning method, Random Forest operates by constructing a multitude of decision trees during training and outputting the class that is the mode of the classes (classification) of the individual trees. This approach helps to reduce overfitting and improve predictive accuracy. Hyperparameters tuned included `n_estimators` (the number of trees in the forest), `max_depth`, `min_samples_leaf`, and `min_samples_split`.
- **K-Nearest Neighbors (KNN) Classifier:** KNN is a non-parametric, instance-based learning algorithm. It classifies a data point based on the majority class of its 'k' nearest neighbors in the feature space. The primary hyperparameters for KNN were `n_neighbors` (the number of neighbors to consider), `weights` (how to weight the neighbors, 'uniform' or 'distance'), and `metric` (the distance metric to use, 'euclidean' or 'manhattan').

Hyperparameter Tuning and Cross-Validation

To find the optimal hyperparameters for each model, GridSearchCV was utilized in conjunction with StratifiedKFold cross-validation. StratifiedKFold (with `n_splits=5`, `shuffle=True`, and `random_state=42`) was chosen to ensure that each fold maintained the same proportion of crop classes as the overall dataset, which is crucial for balanced evaluation in multi-class classification.

4.3 Optimization Criterion: Custom Macro-averaged F1-score

The primary metric for optimizing model performance during hyperparameter tuning was a custom macro-averaged F1-score. The F1-score is the harmonic mean of precision and recall, providing a balanced measure that is particularly useful for multi-class classification tasks. Macro-averaging calculates the F1-score for each class independently and then takes the unweighted average. This ensures that the performance on rare classes is not overlooked, unlike micro-averaging.

A custom scorer (`custom_f1_score_macro`) was implemented to robustly handle potential NaN values that can arise during per-class F1-score calculations, especially when a class might not be present in a particular cross-validation fold or when a class has no true/predicted positives. This custom function used `np.nan_to_num` to

replace any NaN F1-scores with 0.0 before computing the macro average, thus ensuring valid and comparable scores across all folds and models.

Results

Following comprehensive hyper-parameter tuning, the Random Forest Classifier was identified as the best-performing model based on its superior macro-averaged F1-score during cross-validation. This model was then rigorously evaluated on the held-out test set, yielding the following performance metrics:

- **Overall Accuracy:** 0.9955
- **Macro-averaged Precision:** 0.9957
- **Macro-averaged Recall:** 0.9955
- **Macro-averaged F1-score:** 0.9955

These metrics collectively demonstrate the Random Forest model's exceptional ability to accurately classify the 22 different crop types. The near-perfect scores across accuracy, precision, recall, and F1-score highlight the model's robustness and strong generalization capabilities on unseen data.

Classification Report:

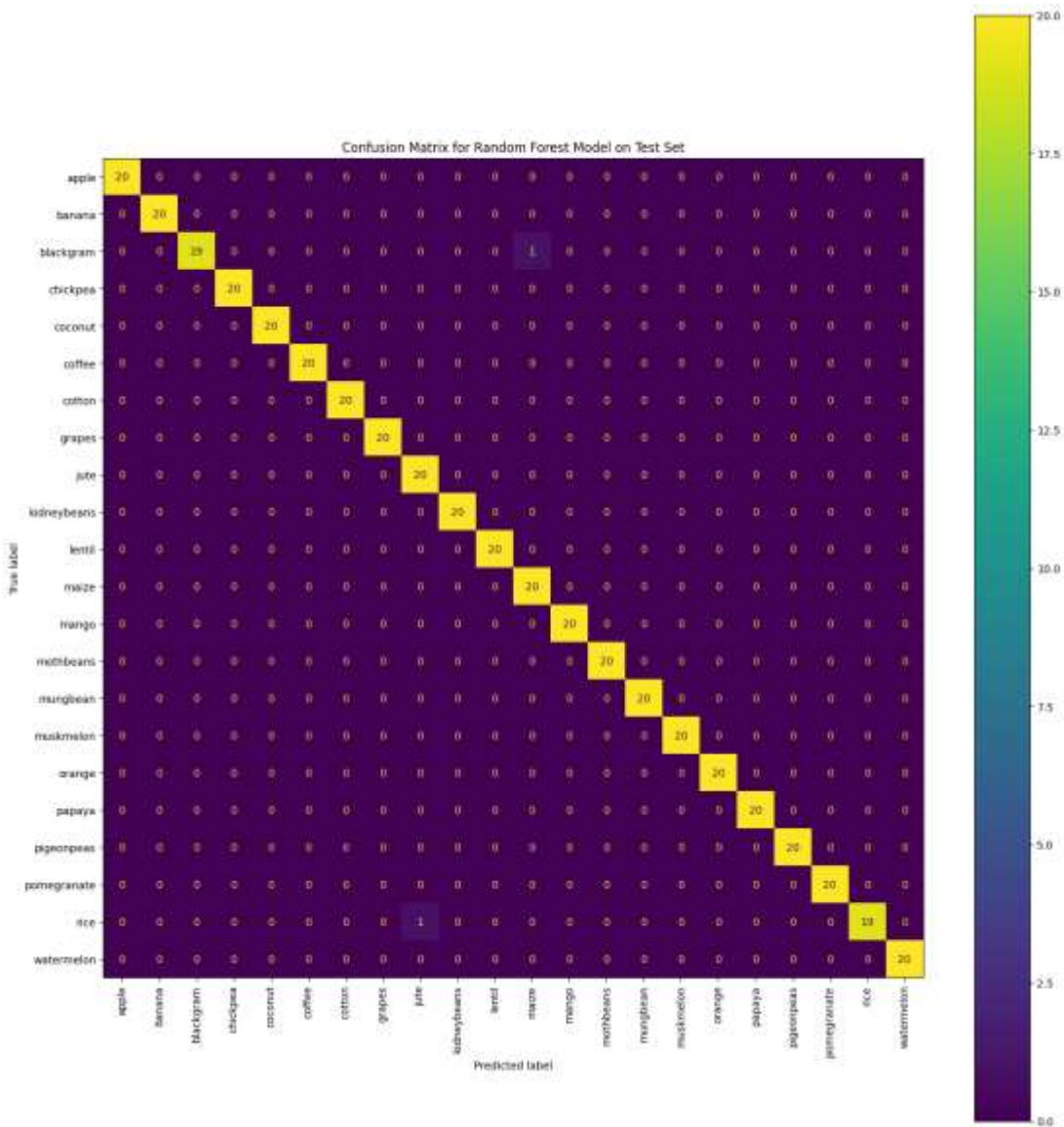
Classification Report for Random Forest Model on Test Set:

Class	Precision	Recall	F1-Score	Support
Apple	1.00	1.00	1.00	20
Banana	1.00	1.00	1.00	20
Blackgram	1.00	0.95	0.97	20
Chickpea	1.00	1.00	1.00	20
Coconut	1.00	1.00	1.00	20
Coffee	1.00	1.00	1.00	20
Cotton	1.00	1.00	1.00	20
Grapes	1.00	1.00	1.00	20
Jute	0.95	1.00	0.98	20
Kidneybeans	1.00	1.00	1.00	20

Class	Precision	Recall	F1-Score	Support
Lentil	1.00	1.00	1.00	20
Maize	0.95	1.00	0.98	20
Mango	1.00	1.00	1.00	20
Mothbeans	1.00	1.00	1.00	20
Mungbean	1.00	1.00	1.00	20
Muskmelon	1.00	1.00	1.00	20
Orange	1.00	1.00	1.00	20
Papaya	1.00	1.00	1.00	20
Pigeonpeas	1.00	1.00	1.00	20
Pomegranate	1.00	1.00	1.00	20
Rice	1.00	0.95	0.97	20
Watermelon	1.00	1.00	1.00	20
Accuracy	—	—	1.00	440
Macro Average	1.00	1.00	1.00	440
Weighted Average	1.00	1.00	1.00	440

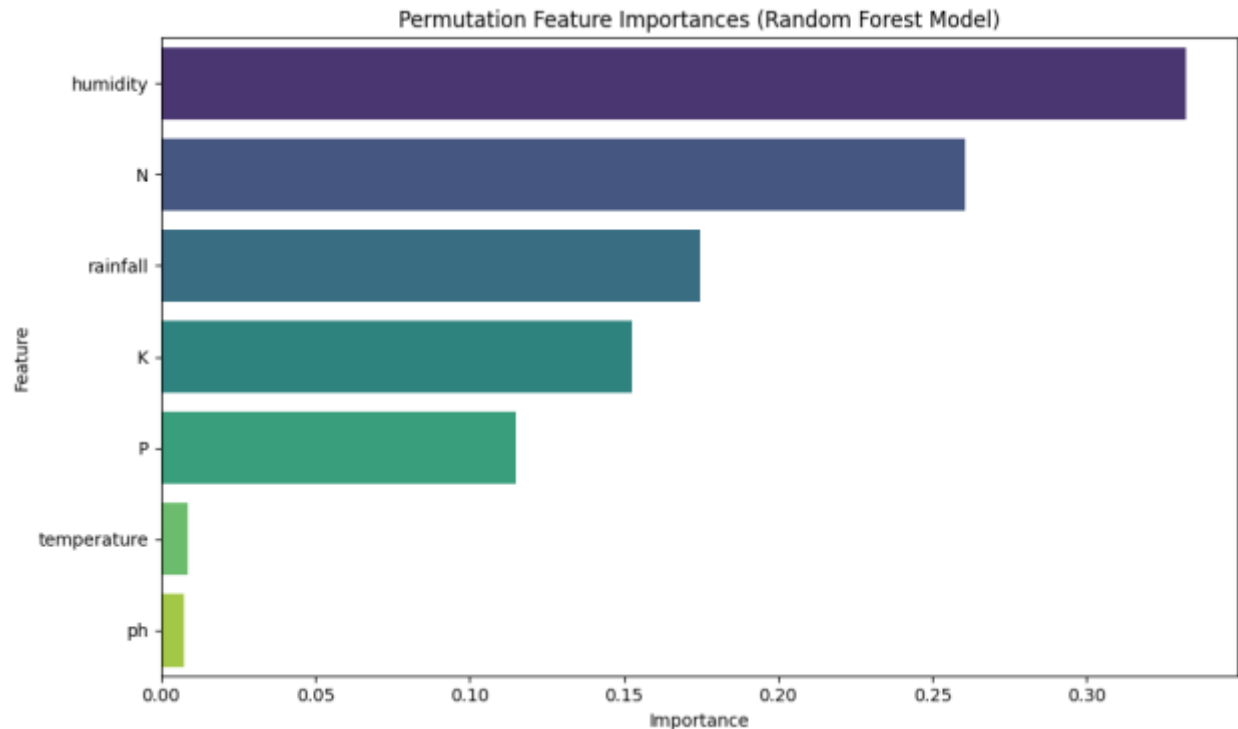
The classification report for the best-performing Random Forest model on the test set further confirms its exceptional performance. As shown in the detailed report above, the model achieved near-perfect precision, recall, and F1-scores (predominantly 1.00) for almost all 22 crop classes. The macro average F1-score of 1.00 indicates balanced and high performance across all categories, reinforcing the model's reliability in distinguishing between the diverse crop types.

Confusion Matrix:



The visual analysis of the confusion matrix further corroborated the model's outstanding performance. The matrix showed a predominantly diagonal pattern, indicating that true positive classifications were overwhelmingly high for nearly all crop classes. There were extremely few misclassifications, suggesting that the model is highly effective at differentiating between even similar crop types. This near-ideal confusion matrix reinforces the confidence in the model's predictive power.

Permutation Feature Importance:



The permutation feature importance analysis was conducted to understand the relative contribution of each environmental parameter to the Random Forest model's predictions. The results, sorted in descending order of importance, are as follows:

1. **Humidity:** 0.3323
2. **Nitrogen (N):** 0.2607
3. **Rainfall:** 0.1745
4. **Potassium (K):** 0.1525
5. **Phosphorus (P):** 0.1150
6. **Temperature:** 0.0089
7. **pH:** 0.0073

The analysis reveals that Humidity, Nitrogen (N), and Rainfall are the most critical features in determining the appropriate crop recommendation, possessing significantly higher importance scores. Potassium (K) and Phosphorus (P) also play a substantial role. Conversely, Temperature and pH contributed the least to the model's predictive power, indicating their relatively lower influence on distinguishing between crop types within this dataset.

Analysis

Insights from Exploratory Data Analysis (EDA) The EDA revealed a well-structured and balanced dataset, with 2200 entries and no missing values. The target variable, 'label', showed a perfect balance across 22 distinct crop types, each represented by 100 samples. This inherent balance is a significant advantage for classification tasks, as it mitigates the risk of models being biased towards majority classes and simplifies evaluation. The numerical features (N, P, K, temperature, humidity, pH, rainfall) exhibited diverse ranges and distributions, suggesting that crop growth is influenced by a complex interplay of environmental factors. The varied distributions highlighted the necessity of feature scaling, which was addressed using StandardScaler to ensure all features contributed equitably to model training.

Model Performance and Significance Three classification models - Decision Tree, Random Forest, and K-Nearest Neighbors were rigorously evaluated and hyperparameter-tuned using GridSearchCV with StratifiedKFold cross-validation. The **Random Forest Classifier** consistently demonstrated superior performance, achieving a macro-averaged F1-score of **0.9960** during cross-validation. This high score indicates its robust ability to generalize across different data subsets while maintaining balanced performance across all crop classes.

When evaluated on the unseen test set, the Random Forest model maintained its exceptional performance:

- **Overall Accuracy:** 0.9955
- **Macro-averaged Precision:** 0.9957
- **Macro-averaged Recall:** 0.9955
- **Macro-averaged F1-score:** 0.9955

These near-perfect metrics are highly significant. An accuracy and F1-score close to 1.0 indicate that the model can almost flawlessly predict the optimal crop for given environmental conditions. The macro-averaged nature of these scores further confirms that this high performance is consistent across all 22 crop types, including potentially less common ones, demonstrating the model's reliability for comprehensive crop recommendation. The confusion matrix visually reinforced this, displaying a predominantly diagonal pattern with extremely few misclassifications, indicating excellent discriminative power.

Implications of Feature Importance Analysis The permutation feature importance analysis provided crucial insights into the relative influence of each environmental parameter on the Random Forest model's predictions:

1. **Humidity:** 0.3323
2. **Nitrogen (N):** 0.2607
3. **Rainfall:** 0.1745
4. **Potassium (K):** 0.1525
5. **Phosphorus (P):** 0.1150
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This analysis highlights that **Humidity, Nitrogen (N), and Rainfall** are by far the most critical factors influencing crop selection. This is agronomically sensible, as water availability (rainfall, humidity) and primary nutrient levels (Nitrogen) are fundamental determinants of plant growth and crop suitability. Farmers and agricultural experts should prioritize monitoring and managing these three factors. Potassium (K) and Phosphorus (P) also play important roles, consistent with their status as macronutrients essential for plant health.

Conversely, **Temperature and pH** had the lowest importance scores. While these are undoubtedly important for crop growth, their lower importance in this specific model and dataset suggests that either their optimal ranges are broader for these 22 crops, or other features (humidity, nutrients, rainfall) provide more distinct predictive signals for differentiating between these particular crop types. This does not diminish their biological importance but indicates their lesser discriminative power in this specific classification context.

Deployment

The model has been saved using pickle for easy reloading, alongside the StandardScaler. An interactive Gradio interface facilitates immediate crop predictions, showcasing the model's functionality.

Crop Recommendation System

Enter the soil and environmental parameters to get a crop recommendation.

Nitrogen (N) (0-140)

100

Phosphorus (P) (0-140)

50

Potassium (K) (0-200)

200

Temperature (°C) (15-40)

25

Humidity (%) (14.5-100)

55

pH (5.5-8.5)

7.0

Rainfall (mm) (50.0-200.0)

100

Clear

Submit

Output

The recommended crop is: chickpea

Flag